AP INTERFACE DESIGN DOCUMENT

Legacy Oracle Purchasing (PO)

→

New Oracle Accounts Payable (AP)

Receipt-Driven AP Invoice Interface

Document Version	1.1
Platform	Oracle EBS R12.2
Interface Type	Cross-Instance DB Link + AP Open Interface
PO Types Covered	Standard, Blanket, Planned, Services
Date	March 11, 2026
Status	DRAFT

The PO module drives supplier invoice processing in AP

For the Legacy Purchasing (PO) to New Oracle AP interface, the Receipt/Receiving process in the legacy PO module shall be the trigger for the required information to be brought to new AP module to create the Oracle AP Invoice using the AP Invoice Interface table.

If the material or services have not been received, then there should be no information interfaced to the new AP module

Return to Vendor (RTV) shall also be addressed – to be discussed with BAMUL team for triggers and process

1. Overview

1.1 Purpose

This Interface Design Document (IDD) describes the technical design and implementation approach for the Receipt-Driven AP Invoice Interface between the Legacy Oracle Purchasing (PO) module (Oracle EBS R12.2) and the new Oracle Accounts Payable (AP) module (Oracle EBS R12.2) running in a separate instance.

1.2 Core Business Rule

NO RECEIPT = NO AP INVOICE

Only purchase order receipts that have been physically confirmed in the legacy Oracle Receiving sub-ledger will trigger the creation of an AP Invoice in the new Oracle AP module. If goods or services have not been received and confirmed, no data will be interfaced to the new AP instance.

1.3 Interface Scope

Attribute	Legacy (Source)	New Instance (Target)
Oracle Instance	LEGACY_INSTANCE	NEW_INSTANCE
Oracle Version	Oracle EBS R12.2	Oracle EBS R12.2
Module	Oracle Purchasing (PO) / Receiving (RCV)	Oracle Accounts Payable (AP)
PO Types	Standard, Blanket, Planned, Services	Standard AP Invoices (STANDARD type)
Trigger Event	RECEIVE transaction confirmed in RCV	Invoice created via AP Open Interface Import

Return to Vendor (RTV) transactions — handled separately in Section 8 (page 11 onwards)

1.4 Out of Scope

- AP Invoices that are not PO-matched (non-PO invoices)
- Invoice approval workflow configuration in new AP
- Payment processing in new AP
- Intercompany PO receipts (separate interface design required)
- Legacy AP invoices already matched and paid prior to go-live cutoff

2. Interface Architecture

2.1 Interface Flow

The interface uses a five-stage Extract-Validate-Transform-Load-Reconcile (EVTLR) pattern:

Step	Stage	Description
1	**EXTRACT**	SQL cursor queries RCV_TRANSACTIONS and related PO tables in the legacy instance via a database link. Only RECEIVE transaction type records with positive received quantity or amount are selected.
2	**VALIDATE**	Each receipt record is validated against business rules: positive receipt quantity/amount, active supplier in new AP, duplicate receipt check, and valid COA mapping for the PO distribution account.
3	**TRANSFORM**	Receipt data is mapped to AP Interface table columns. Legacy Chart of Accounts (CCID) values are cross-walked to new COA via the XXCUST_COA_MAPPING table. Invoice numbers are generated per defined naming convention.
4	**LOAD**	Transformed data is inserted into AP_INVOICES_INTERFACE and AP_INVOICE_LINES_INTERFACE tables in the new AP instance. All records are committed with a unique Run ID and Group ID.
5	**RECONCILE**	After the Payables Open Interface Import program is run, a reconciliation query compares the interface log against AP_INVOICES_ALL to confirm all loaded records were successfully converted to AP invoices.

2.2 Technical Architecture

Component	Detail
Integration Method	Oracle Database Link (DB Link) + Oracle AP Open Interface tables
DB Link Name	LEGACY_INSTANCE (created on new instance, pointing to legacy instance)
Interface Object	PL/SQL Package: XXCUST_PO_AP_INTERFACE_PKG (deployed in new instance)
Target Tables	AP_INVOICES_INTERFACE, AP_INVOICE_LINES_INTERFACE (new instance)
Control Table	XXCUST_PO_AP_INTERFACE_LOG (new instance) — tracks all processed, rejected, and errored records
COA Mapping Table	XXCUST_COA_MAPPING (new instance) — maps legacy CCID to new Chart of Accounts CCID
AP Import Program	Standard Oracle: Payables Open Interface Import (Source: XXCUST_PO_RECEIPT)
Error Handling	Record-level: failed records logged to XXCUST_PO_AP_INTERFACE_LOG with rejection reason; processing continues for remaining records

3. Source-to-Target Column Mapping

3.1 AP Invoice Header — AP_INVOICES_INTERFACE

The following table documents the column-level mapping from legacy PO/RCV source tables to the AP_INVOICES_INTERFACE and AP_INVOICE_LINES_INTERFACE target tables in the new Oracle AP instance.

Target Column	Target Table	Source Table.Column	Transformation Rule
invoice_id	AP_INVOICES_INTERFACE	AP_INVOICES_INTERFACE_S.NEXTVAL	New sequence value — unique per invoice header
invoice_num	AP_INVOICES_INTERFACE	Derived	RCPT-{receipt_num}-L{po_line_num} (zero-padded 3 digits)
invoice_type_lookup_code	AP_INVOICES_INTERFACE	Hardcoded	STANDARD
invoice_date	AP_INVOICES_INTERFACE	SYSDATE	Date interface is run; adjust to receipt date if required
vendor_id	AP_INVOICES_INTERFACE	PO_HEADERS_ALL.vendor_id	Must match vendor_id in new AP_SUPPLIERS (pre-migrated)
vendor_site_id	AP_INVOICES_INTERFACE	PO_HEADERS_ALL.vendor_site_id	Must match vendor_site_id in new AP_SUPPLIER_SITES_ALL
invoice_amount	AP_INVOICES_INTERFACE	RCV_TRANSACTIONS.quantity * po_unit_price	For Services (TEMP LABOR/FIXED PRICE): use amount_received
invoice_currency_code	AP_INVOICES_INTERFACE	PO_HEADERS_ALL.currency_code	Direct copy from PO currency
exchange_rate	AP_INVOICES_INTERFACE	RCV_TRANSACTIONS.currency_conversion_rate	Direct copy; null if functional currency
goods_received_date	AP_INVOICES_INTERFACE	RCV_TRANSACTIONS.transaction_date	Actual receipt confirmation date from legacy
receipt_verified_flag	AP_INVOICES_INTERFACE	Hardcoded	Y — receipt has been confirmed
po_number	AP_INVOICES_INTERFACE	PO_HEADERS_ALL.segment1	PO reference for three-way match in new AP
source	AP_INVOICES_INTERFACE	Hardcoded	XXCUST_PO_RECEIPT — identifies this interface
group_id	AP_INVOICES_INTERFACE	AP_INVOICES_INTERFACE_S.NEXTVAL	Unique batch group ID per interface run
org_id	AP_INVOICES_INTERFACE	PO_HEADERS_ALL.org_id	Operating unit — must be valid in new instance
invoice_line_id	AP_INVOICE_LINES_INTERFACE	AP_INVOICE_LINES_INTERFACE_S.NEXTVAL	New sequence value — unique per invoice line
line_number	AP_INVOICE_LINES_INTERFACE	PO_LINES_ALL.line_num	PO line number used as invoice line number
line_type_lookup_code	AP_INVOICE_LINES_INTERFACE	Hardcoded	ITEM for all goods and services lines
amount	AP_INVOICE_LINES_INTERFACE	RCV_TRANSACTIONS.quantity * po_unit_price	Services: use PO_LINE_LOCATIONS_ALL.amount_received
quantity_invoiced	AP_INVOICE_LINES_INTERFACE	RCV_TRANSACTIONS.quantity	Null for Services (amount-based); populated for Goods

unit_price	AP_INVOICE_LINES_INTERFACE	RCV_TRANSACTIONS.po_unit_price	Null for Services; populated for Goods
unit_meas_lookup_code	AP_INVOICE_LINES_INTERFACE	RCV_TRANSACTIONS.unit_of_measure	Direct copy from receipt UOM
po_header_id	AP_INVOICE_LINES_INTERFACE	PO_HEADERS_ALL.po_header_id	PO reference for matching in AP
rcv_transaction_id	AP_INVOICE_LINES_INTERFACE	RCV_TRANSACTIONS.transaction_id	Receipt transaction reference for three-way match
dist_code_combination_id	AP_INVOICE_LINES_INTERFACE	XXCUST_COA_MAPPING.new_ccid	New COA CCID looked up from legacy CCID via mapping table
description	AP_INVOICE_LINES_INTERFACE	PO_LINES_ALL.item_description	Concatenated: item description + PO line + receipt date
org_id	AP_INVOICE_LINES_INTERFACE	PO_HEADERS_ALL.org_id	Operating unit — must be valid in new instance

4. Business Rules and Validation

4.1 Receipt Qualification Rules

The following rules determine whether a receipt transaction qualifies to generate an AP Invoice:

#	Rule	Logic	Action if Failed
BR -01	**RECEIVE Transaction Type Only**	RCV_TRANSACTIONS.transaction_type = 'RECEIVE'. Excludes RETURN, CORRECT, DELIVER, TRANSFER and all other transaction types.	Exclude record. Do not interface.
BR -02	**Positive Received Quantity or Amount**	quantity > 0 for Goods POs, OR amount_received > 0 for Service POs. Zero or negative receipts are excluded.	Reject with reason. Log to XXCUST_PO_AP_INTERFACE_LOG.
BR -03	**PO Source Document Only**	RCV_TRANSACTIONS.source_document_code = 'PO'. Excludes internal order receipts, RMA receipts, and other source types.	Exclude record. Do not interface.
BR -04	**Active PO — Not Cancelled or Finally Closed**	PO_HEADERS_ALL.cancel_flag = 'N' AND closed_code NOT IN ('FINALLY CLOSED','CANCELLED'). PO_LINE_LOCATIONS_ALL.cancel_flag = 'N'.	Reject with reason. Log to XXCUST_PO_AP_INTERFACE_LOG.
BR -05	**Supplier Must Exist in New AP Instance**	vendor_id must be found in AP_SUPPLIERS in the new AP instance. Supplier master migration is a prerequisite for this interface.	Reject with reason. Supplier must be migrated before re-running.
BR -06	**No Duplicate Processing**	RCV_TRANSACTIONS.transaction_id must not already exist in XXCUST_PO_AP_INTERFACE_LOG with status = PROCESSED. Prevents double-invoicing of the same receipt.	Reject with reason: 'Duplicate — already processed'.
BR -07	**COA Mapping Must Exist**	PO_DISTRIBUTIONS_ALL.code_combination_id must have a corresponding active record in XXCUST_COA_MAPPING. The new CCID (new_ccid) must be non-null.	Reject with reason. COA mapping table must be updated before re-running.
BR -08	**Quantity Billed Check**	PO_DISTRIBUTIONS_ALL.quantity_billed must be less than RCV_TRANSACTIONS.quantity. Fully billed receipts are excluded to prevent over-invoicing.	Exclude record. Do not interface.

4.2 PO Type Handling

PO Type	Purchase Basis	Interface Logic
Standard PO	GOODS	Invoice amount = quantity_received × po_unit_price. quantity_invoiced and unit_price populated on invoice line.
Blanket PO Release	GOODS	Same as Standard PO. Blanket release shipment is referenced via po_line_location_id.
Planned PO Release	GOODS	Same as Standard PO. Planned release treated equivalently.
Services PO (Fixed Price)	FIXED PRICE	Invoice amount = PO_LINE_LOCATIONS_ALL.amount_received.

		quantity_invoiced and unit_price are NULL on the invoice line (amount-based invoicing).
Services PO (Time & Material)	TEMP LABOR	Invoice amount = PO_LINE_LOCATIONS_ALL.amount_received. quantity_invoiced and unit_price are NULL on the invoice line.

5. Error Handling and Logging

5.1 Error Handling Strategy

The interface implements record-level error handling. An error on an individual receipt record does not stop processing of subsequent records. All errors, rejections, and successfully processed records are written to the custom log table XXCUST_PO_AP_INTERFACE_LOG.

Status	Meaning	Resolution
PROCESSED	Successfully loaded to AP Interface tables	Run Payables Open Interface Import in new AP to complete the invoice creation.
REJECTED	Failed one or more business rule validations	Review rejection_reason column. Fix underlying data issue (e.g. add COA mapping, migrate supplier) and re-run the interface for affected records.
ERROR	Unexpected technical/system error during processing	Review rejection_reason for SQLERRM details. Investigate data or DB link connectivity issue. Re-run after fix.

5.2 AP Open Interface Import Error Handling

After the PL/SQL interface loads data into AP_INVOICES_INTERFACE, the standard Oracle Payables Open Interface Import program processes the records. If AP Import rejects records, they remain in the interface table with a STATUS value set by Oracle AP. Common AP Import rejection reasons include:

- Invalid or inactive vendor/vendor site in new AP instance
- Invalid distribution account CCID in new GL Chart of Accounts
- Duplicate invoice number for the same vendor
- Invoice amount does not balance to line amounts
- Invalid operating unit or org_id for the new instance

To review AP Import rejections, query the interface tables after the import run:

```
SELECT invoice_num, vendor_id, invoice_amount, status, reject_lookup_code FROM
AP_INVOICES_INTERFACE WHERE source = 'XXCUST_PO_RECEIPT' AND status IS NOT NULL;
```

6. Prerequisites and Execution Steps

6.1 Prerequisites

The following must be completed before the interface can be executed:

#	Prerequisite	Detail
P-01	**Supplier Master Migration**	All legacy suppliers (AP_SUPPLIERS) and supplier sites (AP_SUPPLIER_SITES_ALL) must be migrated to the new AP instance prior to running this interface. Vendor IDs must be preserved across instances, or a vendor ID cross-reference table must be implemented.
P-02	**Database Link Creation**	A database link named LEGACY_INSTANCE must be created on the new Oracle instance pointing to the legacy instance. SELECT privileges must be granted on all source tables referenced in the interface cursor.
P-03	**COA Mapping Table Populated**	XXCUST_COA_MAPPING must be populated with all legacy GL code combination IDs (CCIDs) and their corresponding new Chart of Accounts segments and new CCID values before any receipts with those accounts can be interfaced.
P-04	**Supporting Tables Created**	DDL statements in Section 1 of the PL/SQL script (XXCUST_PO_AP_INTERFACE_LOG and XXCUST_COA_MAPPING) must be executed in the new instance.
P-05	**Package Compiled**	XXCUST_PO_AP_INTERFACE_PKG package specification and body must be compiled without errors in the new Oracle AP instance.
P-06	**AP Interface Source Registered**	The source name XXCUST_PO_RECEIPT must be registered in AP_LOOKUP_CODES with lookup type SOURCE in the new AP instance, or use an existing source code.

6.2 Execution Steps

Step	Action	Detail
1	**Populate COA Mapping**	Insert all required legacy CCID to new CCID mappings into XXCUST_COA_MAPPING. Validate new_ccid values are valid in the new GL instance.
2	**Run PL/SQL Interface**	Execute XXCUST_PO_AP_INTERFACE_PKG.run_interface() with desired parameters: operating unit, receipt date range, and optional single PO filter. Note the Run ID from the output log.
3	**Review Interface Log**	Query XXCUST_PO_AP_INTERFACE_LOG for the Run ID. Investigate and resolve any REJECTED or ERROR records before proceeding.
4	**Run AP Open Interface Import**	In the new AP instance, navigate to: Payables Manager > Other > Import > Invoices. Set Source = XXCUST_PO_RECEIPT. Set Group ID from the interface run output.
5	**Run Reconciliation Query**	Execute the reconciliation query (Section 3 of SQL script) with the Run ID. Confirm all PROCESSED interface records have a corresponding AP Invoice in AP_INVOICES_ALL.
6	**Resolve AP Import Rejections**	For any AP Import rejections, fix the underlying data issue, delete the rejected record from AP_INVOICES_INTERFACE, and re-load the specific record.

7. Reconciliation

7.1 Reconciliation Approach

A three-level reconciliation must be performed after each interface run to ensure data integrity between the legacy PO/Receiving module and the new Oracle AP module.

#	Check	Description	Source
R-01	**Receipt Count vs Interface Log Count**	Total RECEIVE transactions in legacy RCV_TRANSACTIONS for the date range must equal PROCESSED + REJECTED records in XXCUST_PO_AP_INTERFACE_LOG.	XXCUST_PO_AP_INTERFACE_LOG
R-02	**Interface Log vs AP Interface Table**	All records with status = PROCESSED in XXCUST_PO_AP_INTERFACE_LOG must have a corresponding record in AP_INVOICES_INTERFACE (before AP Import run).	AP_INVOICES_INTERFACE
R-03	**AP Interface Table vs AP Invoices**	After AP Open Interface Import, all records in AP_INVOICES_INTERFACE with source = XXCUST_PO_RECEIPT must have corresponding records in AP_INVOICES_ALL.	AP_INVOICES_ALL
R-04	**Total Amount Reconciliation**	Sum of receipt_amount in XXCUST_PO_AP_INTERFACE_LOG (PROCESSED status) must equal sum of invoice_amount in AP_INVOICES_ALL for the same batch.	Both tables

8. Return to Vendor (RTV) Transaction Handling

8.1 Overview

Return to Vendor (RTV) transactions in the legacy Oracle Receiving module represent the reversal of a previously confirmed receipt. Because this interface is receipt-driven, RTVs must be handled with equal rigour to ensure the new Oracle AP module accurately reflects only the net goods and services actually received and retained.

RTV transactions are recorded in RCV_TRANSACTIONS with transaction_type = 'RETURN TO VENDOR'. They reference the same PO line location as the original RECEIVE transaction and carry a negative or equal quantity to the original receipt.

8.2 RTV Scenarios

Three distinct scenarios govern how RTVs interact with this interface, depending on the timing of the return relative to the interface run:

Ref	Scenario	Description	Interface Action
RTV-A	**RTV before AP Invoice is created**	Goods are received and then returned to the vendor before the interface run executes and before any AP Invoice is created in the new instance.	Net quantity check prevents an AP Invoice from being created. If net received quantity = 0, no invoice is interfaced. If partial return, invoice is created for net received quantity only.
RTV-B	**Full RTV after AP Invoice already created**	The receipt was already interfaced and an AP Invoice exists in new AP. All goods are subsequently returned to the vendor in full.	A Credit Memo for the full invoice amount must be raised in new AP to fully reverse the original invoice. This requires a separate RTV-to-Credit-Memo interface run.
RTV-C	**Partial RTV after AP Invoice already created**	The receipt was already interfaced and an AP Invoice exists in new AP. Only a portion of the received goods are subsequently returned.	A partial Credit Memo (returned_quantity x unit_price) must be raised in new AP. The original invoice remains open for the retained quantity.

8.3 Scenario RTV-A: Net Quantity Logic (Enhancement to Receipt Cursor)

To implement Scenario RTV-A, the receipt extraction cursor in XXCUST_PO_AP_INTERFACE_PKG must be enhanced to net off any RTV quantities before deciding whether to create an AP Invoice. The following logic is applied per PO line location:

```
Net Receivable Quantity = SUM(RECEIVE qty) - SUM(RETURN TO VENDOR qty)
```

Additionally, the invoice amount loaded to AP_INVOICES_INTERFACE uses the net received quantity rather than the raw RECEIVE transaction quantity:

```
invoice_amount = net_received_quantity x po_unit_price
```

8.4 Scenario RTV-B and RTV-C: RTV-to-Credit-Memo Interface

Where an AP Invoice already exists in the new instance and a subsequent RTV is recorded in the legacy Receiving module, a Credit Memo must be created in new AP. This is handled by a dedicated RTV-to-Credit-Memo interface which mirrors the receipt-to-invoice interface but operates in reverse.

Attribute	Detail
Trigger	RCV_TRANSACTIONS.transaction_type = 'RETURN TO VENDOR' in legacy instance
Source Tables	RCV_TRANSACTIONS, RCV_SHIPMENT_HEADERS, PO_HEADERS_ALL, PO_LINES_ALL, PO_LINE_LOCATIONS_ALL, PO_DISTRIBUTIONS_ALL
Target Tables	AP_INVOICES_INTERFACE (invoice_type_lookup_code = 'CREDIT'), AP_INVOICE_LINES_INTERFACE
Invoice Type	CREDIT — this creates a Credit Memo in Oracle AP linked back to the original Standard invoice
Credit Amount	RTV-B (Full): full original invoice amount. RTV-C (Partial): returned_quantity x po_unit_price
Invoice Number Convention	CM-RCPT-{receipt_number}-L{po_line_num} — prefixed with CM to distinguish from original invoice
Duplicate Prevention	RTV transaction_id checked against XXCUST_PO_AP_INTERFACE_LOG before loading — same duplicate-check logic as receipt interface
COA Mapping	Same XXCUST_COA_MAPPING table used — legacy CCID from PO distribution mapped to new CCID for credit memo distribution line

8.5 RTV Business Rules

Rule	Description	Logic	Action if Failed
RTV-BR-01	**RETURN TO VENDOR Transaction Type Only**	RCV_TRANSACTIONS.transaction_type = 'RETURN TO VENDOR'. No other transaction types processed by the RTV interface.	Exclude. Do not create Credit Memo.
RTV-BR-02	**Returned Quantity Must Be Positive**	RCV_TRANSACTIONS.quantity > 0. RTVs are stored as positive quantities in Oracle Receiving (the transaction type implies the direction).	Reject with reason. Log to interface log.
RTV-BR-03	**Original Receipt Must Have Been Interfaced**	For RTV-B and RTV-C: the original RECEIVE transaction_id must exist in XXCUST_PO_AP_INTERFACE_LOG with status = PROCESSED. If not found, no Credit Memo is required (Scenario RTV-A applies).	Skip — original invoice was never created; no Credit Memo needed.
RTV-BR-04	**Credit Amount Cannot Exceed Original Invoice Amount**	returned_quantity x unit_price must not exceed the original interfaced invoice amount for the same PO line. Prevents over-crediting.	Reject with reason. Manual review required.
RTV-BR-05	**No Duplicate RTV Processing**	RCV_TRANSACTIONS.transaction_id (for the RTV record) must not already exist in XXCUST_PO_AP_INTERFACE_LOG with status = PROCESSED.	Reject with reason: Duplicate RTV already processed.

8.6 RTV Reconciliation

After the RTV-to-Credit-Memo interface runs and the Payables Open Interface Import is executed, the following reconciliation checks must be performed:

#	Check	Description

R-05	**RTV Count vs Credit Memo Count**	Total RETURN TO VENDOR transactions in legacy RCV_TRANSACTIONS for the period must equal Credit Memo records created in AP_INVOICES_ALL (invoice_type = CREDIT) for source XXCUST_PO_RECEIPT.
R-06	**Net AP Balance per PO Line**	For each PO line: (Original Invoice Amount) - (Credit Memo Amount) must equal (Net Received Quantity x Unit Price). This confirms the AP balance mirrors the physical net receipts in the legacy Receiving module.
R-07	**Zero-Balance PO Lines**	PO lines where total received quantity equals total returned quantity (full RTV) must have zero net AP exposure — i.e. the Credit Memo fully offsets the original invoice.

9. Document Change History

Version	Date	Author	Changes
1.0	Initial release	Interface Design	Initial document — Receipt-Driven AP Invoice Interface covering all PO types on Oracle EBS R12.2
1.1	Mar 11, 2026	Interface Design	Added Section 8: Return to Vendor (RTV) handling — three RTV scenarios, net quantity logic, RTV-to-Credit-Memo interface design, RTV business rules, and RTV reconciliation checks.